

cooling that the researchers say works as nicely as water cooling. The technique hinges around the fact, as usual, that nanotubes are small. Ions (electrically-charged atoms) are generated using electrodes separated by as little as 10 microns. The cathode is made of nanotubes with tips just as wide as 5 nm.

The ionisation of the air leads to an imbalance in the charges on the electrodes, which results in "lightning bolts." This nano-lightning is responsible for the tiny breezes that do the cooling.

Normally, lightning requires thousands of volts, but here it's been done with a hundred volts and less—because the nanotube tips are so narrow, and because the electrodes are so close together.

These three examples aside, there are many people doing different kinds of work on cooling using nanotubes, and nanotube cooling could be one of the first major "solutions to a real-world problem" that nanotech rolls out.

Also Good As Interconnects

Now we get into our "As chips get smaller and faster" mode of speaking. And so, as chips get smaller and faster, copper interconnects get more difficult and expensive to fabricate; their electrical properties degrade at the scales we're approaching.

Electromigration is the name for the phenomenon whereby the reliability of integrated circuits degrades—because it makes nanometre-sized copper interconnects unreliable, and can even lead to wire failure. Visit <http://en.wikipedia.org/wiki/Electromigration> for an explanation of the phenomenon.

Finally, at one million amperes per square centimetre, copper just burns out. Enter the nanotube, as usual: they can carry a *thousand* times that much. In addition, bundles of densely packed nanotubes can have significantly lower resistance than copper—and as you're aware, lower resistance is a much sought-after attribute.

In addition, with nanotubes, there is no need to create deep, narrow trenches on silicon wafers in which copper conductors are traditionally buried—further increasing the potential for miniaturisation.

With all these advantages to be had, and with all the difficulties that come in the way of nanotubes being used as interconnects, it's only natural that almost every nanotech organisation—and some non-nanotech companies as well—are in a race to bring out nanotube-interconnect-enabled chips.

Scientists at the NASA Ames Research Center (ARC) are amongst those working on the problem. One of the problems is that of the nanotubes being entangled; when they are, they don't display their fantastic current-carrying prowess. The idea at the ARC is to develop a process to untangle them.

In January of this year, it was reported that researchers at Rensselaer Polytechnic Institute had created hybrid structures that "combine the best properties of carbon nanotubes and metal nanowires." These structures could help overcome some of the main hurdles to using nanotubes in chips, displays, sensors, and such.

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Of particular interest is the fact that the researchers' approach allows "the precise attachment of nanotubes to individual metal pins, offering a practical solution to the problem of using carbon nanotubes as interconnects."

As a final word, Intel, too, is looking at nanotubes as a replacement for copper interconnects. The company has managed to create prototype interconnects out of nanotubes, and measure how well the interconnects perform.

The whole nanotube interconnect thing points, essentially, to what you'd expect: shrinkage. And we are, naturally, going to see a lot of that—just like we've been seeing the past couple of decades—so expect the nanotube to begin taking over a few years from now.

The Rise Of The Nanotube comes down to three things: innovate and think up new uses for carbon nanotubes, given all their wonderful properties; find more of those wonderful properties; and most important as of now, find how best to manufacture and interface them.

Manufacture is notoriously difficult; for example, in one process, they come out as a mixed bundle of conducting and semi-conducting, and it's hard to separate them. Another problem, for example, is how to un-bundle them.

Some of what we've talked about above could just turn out to be that much more thin air; some of it could turn out to be groundbreaking. In any case, look forward to the "nanotube computer," which, by the way, is also being talked about. ■

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